

VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

**(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)**

Department of Computer Engineering



Project Report on

AI - Powered Weather Intelligence for Smart Lighting

In partial fulfillment of the Fourth Year (Semester–VII), Bachelor of Engineering
(B.E.) Degree in Computer Engineering at the University of Mumbai

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Project Mentor

Dr.(Mrs.) Nupur Giri

Submitted by

Soham Parab, D17B/41

Sushanth Shetty, D17B/51

Atharva Sambhaji, D17B/47

Vighnarth Nile, D17B/37

(2025-26)

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CERTIFICATE of Approval

This is to certify that **Soham Parab , Sushanth Shetty , Atharva Sambhaji , Vighnarth Nile** of Fourth Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on “**AI - Powered Weather Intelligence for Smart Lighting**” as a part of the coursework of PROJECT-I for Semester-VII under the guidance of **Dr.(Mrs).Nupur Giri** in the year 2024-2025.

Date

Internal Examiner

External Examiner

Project Mentor

Head of the Department

Dr. Mrs. Nupur Giri

Principal

Dr. J. M. Nair

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR B.E PROJECT

Learners will be to:-

Course Outcome	Description of the Course Outcome
CO 1	Do literature survey/industrial visit and identify the problem of the selected project topic.
CO2	Apply basic engineering fundamental in the domain of practical applications FOR problem identification, formulation and solution
CO 3	Attempt & Design a problem solution in a right approach to complex problems
CO 4	Cultivate the habit of working in a team
CO 5	Correlate the theoretical and experimental/simulations results and draw the proper inferences
CO 6	Demonstrate the knowledge, skills and attitudes of a professional engineer & Prepare report as per the standard guidelines.

ABSTRACT of the project

India and rapidly developing nations face a critical dual challenge—the inefficient energy consumption of urban infrastructure and the need to maintain uncompromised public safety during dynamic environmental events. Conventional street lighting systems rely on fixed schedules or basic ambient light sensors, leading to substantial energy waste during mild conditions and critical safety gaps during visibility-reducing events like dense fog, heavy rainfall, or high traffic congestion. This deficiency highlights the urgent need for a structured, adaptive mechanism to intelligently harmonize energy usage with situational visibility requirements.

Our proposed system, the AI - Powered Weather Intelligence for Smart Lighting, acts as an autonomous digital controller for urban light poles. The platform leverages an AI-driven Multi-Agent Pipeline (LangGraph) to ingest and fuse heterogeneous data from Weather APIs, CCTV Cameras, and IoT Sensors. It utilizes Sensor Fusion to calculate a real-time situational awareness score and incorporates an LLM System Monitor to independently validate all control actions for safety and reliability.

By combining a robust FastAPI/Next.js architecture with advanced LLM reasoning and contextual support via RAG (Retrieval-Augmented Generation), the system automates optimal brightness adjustments, allowing for continuous adaptation. This initiative significantly reduces municipal energy consumption, minimizes operational costs, and enhances urban safety and resilience, contributing directly to national goals for smart city governance and environmental sustainability.

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Chapter 1: Introduction

1.1 Introduction to the Project

Traditional smart city lighting systems are often inefficient, relying on rigid scheduling or basic photocells which fail to adapt to complex and dynamic environmental conditions. This leads to wasted energy during mild conditions and inadequate visibility during critical events like dense fog, heavy rainfall, or high traffic congestion. The Smart Lighting Control System addresses this challenge by establishing a sophisticated, autonomous multi-agent pipeline capable of fusing heterogeneous sensor data (Weather API, CCTV, IoT Sensors) to make proactive, intelligent brightness decisions. The system utilizes modern AI orchestration (LangGraph) and LLMs (Groq Llama 3.1) to ensure energy efficiency and maximum public safety.

1.2 Motivation for the Project

- To achieve substantial energy savings and reduce operational costs by ensuring light poles only operate at the minimum safe brightness required by conditions.
- To significantly improve public safety and traffic flow by automatically increasing visibility during adverse weather (cyclone alerts, heavy fog) or high-risk traffic incidents.
- To harness state-of-the-art AI by building a robust multi-agent architecture (LangGraph) that includes decision validation (System Monitor) and contextual learning (RAG + LLM Insights).
- To centralize oversight through an interactive dashboard, allowing operators to visualize complex, fused data and test scenarios instantly.

1.3 Drawback of the Existing System

- Reactive and Single-Factor Decisions: Current systems are often only reactive to ambient light (single sensor) or rigid time schedules, failing to account for complex visibility factors like fog density, pavement wetness, or wind.
- Data Silos and Poor Correlation: Traditional systems fail to correlate data between disparate city networks (e.g., linking high CCTV traffic congestion with high IoT noise levels) to make an informed, comprehensive decision.

- **Non-Adaptive Logic:** The decision-making process is typically fixed (rules-based) and non-adaptive, leading to suboptimal energy usage when conditions suddenly improve or deteriorate.
- **Lack of Safety Review:** There is no automated, independent check on control actions, risking potential human or algorithmic errors that could compromise safety (e.g., dimming lights during a critical event).

1.4 Problem Definition

How to build a secure, high-reliability, technology-driven multi-agent system that:

1. Integrates, preprocesses, and cleans heterogeneous sensor inputs from the External World (Weather, IoT, CCTV).
2. Automates Sensor Fusion and Anomaly Detection to generate a dynamic, comprehensive Situational Awareness score.
3. Leverages RAG + LLM Insights to generate an adaptive and contextually relevant Decision Engine recommendation.
4. Ensures safety by routing the decision through an independent System Monitor (LLM Judge) for validation before executing the control action on the Smart Streetlights.

1.5 Relevance of the Project

- **Advancing Urban Resilience:** The system automates response to high-impact events (cyclones, severe weather) that pose the greatest threat to city operations and safety.
- **Environmental and Economic Impact:** Directly contributes to the reduction of urban energy consumption and CO2 footprint, aligning with global sustainable development goals.
- **Operator Trust and Auditability:** Provides a fully transparent, auditable trail of every AI decision (including the LLM Judge's final verdict) to enhance operator trust and regulatory compliance.
- **Framework for Future Systems:** The multi-agent LangGraph architecture provides a reusable blueprint for managing other city infrastructure systems (e.g., water, waste, traffic lights).

1.6 Methodology Used

1. **Agent Architecture Design:** Defined a five-node LangGraph Agent Pipeline (Collector, Preprocessing, Fusion, Decision, Monitor) to manage the entire workflow.

2. Sensor Fusion Logic: Developed rules and algorithms to correlate and merge data, calculating composite metrics like a situational awareness score and cross-sensor validation.
3. RAG Implementation: Setup of a Pinecone vector database seeded with historical incidents, queried by LangChain to provide contextual data to the Decision Engine.
4. LLM Integration: Utilized Groq (Llama 3.1) for low-latency AI decision-making (Agent 2) and used a separate call for the critical safety review in the System Monitor (Agent 4).
5. Real-time Frontend: Developed a Next.js dashboard featuring a Leaflet map with WebSocket integration for real-time visualization of pole status and AI control updates.
6. Dynamic UI: Implemented custom SVG icons on the map that dynamically change based on the currently simulated weather condition for enhanced visual clarity.

Chapter 2: Literature Survey

2.1 Research Papers

- **Paper 1: Internet of Things-Based Multi-Agent System for the Control of Smart Street Lighting (2024)**

- a. Abstract of the research paper

- This work presents a framework for intelligent street lighting control utilizing an IoT-based Fuzzy Multi-Agent System (MAS) approach. The solution aims to enhance human safety, reduce CO_2 emissions, and optimize energy consumption by controlling smart street lighting remotely. The MAS models the system as a set of autonomous, intelligent, and cooperative entities, while Fuzzy Logic is adopted to handle the imprecision and uncertainty inherent in multi-sensor data gathering.

- b. Inference drawn from the paper

- The study validates the use of a Multi-Agent System as an appropriate paradigm for modeling the complex, distributed nature of smart street lighting systems. It confirms the project's design choice to use independent agents for control and emphasizes the need for mechanisms (like the project's Preprocessing/Fusion module) to handle imprecision and uncertainty in diverse IoT data streams.

- **Paper 2: Multi-Agent Systems for Adaptive Traffic Light Control in Smart Cities (2025)**

- a. **Abstract of the research paper**

- This paper examines the design, learning, and deployment of **Multi-Agent Systems (MAS)** for adaptive traffic light control, a cornerstone technology for reducing congestion and improving safety in smart cities. It synthesizes architectural patterns for decentralized and hierarchical agent control and discusses the critical need for **data fusion techniques** to reconcile heterogeneous inputs from loop detectors, cameras, and connected vehicles. The research highlights that MAS architecture enhances robustness and scalability in urban environments.

- b. **Inference drawn from the paper**

- The findings directly support the use of MAS to manage city infrastructure, which is inherently distributed and complex. Critically, it confirms that **robust data fusion** is essential to reconcile the project's inputs (Weather, CCTV Traffic, IoT) for reliable situational awareness, justifying the project's **Sensor Fusion** module.

- **Paper 3: Intelligent Streetlight Control System Using Machine Learning Algorithms for Enhanced Energy Optimization in Smart Cities (2025)**

- a. **Abstract of the research paper**

- This research proposes an adaptive **Machine Learning (ML)** system for streetlight control designed to react automatically to the environment and human activity patterns in real time. The objective is to create an integrated system that uses **combined sensor data** from ambient detectors, weather stations, traffic cameras, and occupancy sensors. The paper argues this shift from legacy, fixed-timer systems is necessary to unlock major energy savings and maximize operational autonomy.

- **b. Inference drawn from the paper**

- The study validates the core problem statement: that most municipalities use non-dynamic infrastructure, missing major energy savings. It strongly justifies the project's decision to use **combined sensor data** as the input foundation, proving that ML models (and by extension, LLM agents) reveal complex patterns that standard rule-based systems cannot.

- **Paper 4: Design and Application of a Smart Lighting System Based on Distributed Wireless Sensor Networks (2020)**

- a. Abstract of the research paper This study aims to reduce energy consumption by implementing a smart lighting system that integrates sensor technologies and illumination control rules. The architecture consists of a distributed wireless sensor network (WSN) utilizing ZigBee, with sensing modules that include passive infrared (PIR) and microwave Doppler sensors for occupancy, as well as ambient light sensors. The dimming level of each luminaire is controlled by a ruleset that takes into account both real-time occupancy and daylight harvesting to optimize energy use.

- b. Inference drawn from the paper

- This paper validates the core project goal of using sensor networks and intelligent control rules for energy reduction.
- It demonstrates the effectiveness of a distributed system (like the project's multi-agent model) over a centralized one.
- The study confirms the necessity of integrating multiple sensor types (occupancy, ambient light) to make an informed decision, which aligns with this project's use of Weather, CCTV, and IoT data.

- **Paper 5: Multi-agent system with information fusion for intelligent lighting control (2014)**
- a. Abstract of the research paper This paper proposes a multi-agent system (MAS) for intelligent lighting control, designed to improve user-centric comfort and save energy. A multi-source information fusion model is designed to blend heterogeneous data, including ambient visual information, occupancy data, energy data, and occupant preferences, to enable both automatic and human-controlled decisions. The architecture includes "Sensor agents" for preprocessing raw data and "Local lighting agents" that use the fused data to make reliable estimates and control decisions.
- b. Inference drawn from the paper
 - This paper provides strong academic justification for the project's core architecture, explicitly naming both Multi-Agent Systems and Information Fusion as critical components for intelligent control.
 - It directly supports the project's pipeline structure by suggesting "Sensor agents" for preprocessing data, which maps to the "Data Preprocessing Module".
 - The concept of blending data for "automatic control" and "human control" validates the project's inclusion of both an autonomous Decision Engine and an operator-facing Admin Dashboard for configuration.
 - It confirms that a MAS approach is ideal for handling concurrent data processing and making distributed, real-time decisions.

2.2 Books / Articles / News/Apps

- **Dr. Patel & K. Sharma, "Multi-Agent AI for Urban Resilience: A Review on Smart City Infrastructure Control."**
 - This academic paper addresses the shift from centralized SCADA systems to decentralized, multi-agent architectures for infrastructure control. It provides the

core justification for the project's LangGraph/Groq pipeline by highlighting the superior real-time responsiveness and fault tolerance of autonomous agents [1].

- **Industry Report: "Smart Street Lighting Expected to Cut Energy Consumption by 60% by 2030." (Global Energy Monitor)**
 - This industry forecast underscores the economic and sustainability imperative behind the project. It validates the primary goal of the system: using AI to achieve maximum energy efficiency while maintaining critical safety standards in cities globally [2].
- **White Paper: "Improving Pedestrian Safety with Heterogeneous Sensor Fusion in Urban Lighting."**
 - This technical study demonstrates that combining data from multiple sources (CCTV traffic, weather, AQI) yields better lighting decisions than relying on simple ambient light sensors alone. This directly supports the project's Sensor Fusion + Edge AI module [3].

2.3 Interaction with Domain Experts

Practitioner feedback aligns with documented smart-city roadmaps that emphasize pan-city “smart solutions,” energy-efficient street lighting, and resilience indices; this validates the project’s focus on zone-based fault isolation, rapid failover, and real-time visualization for operations teams.

2.5 Existing Systems / Apps

Upon reviewing the current market, existing solutions for smart lighting control fall into three main categories:

1. **Legacy Timer-Based Systems:** This is the most common and traditional method, where streetlights are controlled by a simple astronomical time clock. They turn on at dusk and off at dawn based on a fixed schedule, with no consideration for real-time weather or environmental conditions.
2. **Basic Sensor-Based (PLC) Systems:** These are first-generation smart lights that use a single sensor, typically a photocell (ambient light sensor), to turn on or off. While more

responsive than timers, they are still "dumb," as they cannot differentiate between a dark, clear night and a dark, foggy afternoon, leading to unsafe or inefficient lighting levels.

3. Siloed "Smart City" Platforms (e.g., CityOS): These are large-scale, commercial management platforms (like those from Siemens, Cisco, or Signify) that provide a dashboard for monitoring city assets. While they can integrate multiple data feeds (like weather or traffic), the data is typically presented on separate, siloed dashboards for a human operator to interpret. The control logic is often proprietary and based on rigid, pre-programmed rules (e.g., IF rain THEN brightness = 70%).

2.6 Lacuna in Existing Systems

The existing systems, while foundational, suffer from several significant gaps (lacunae) that AI - Powered Weather Intelligence for Smart Lighting is designed to address:

- **Lack of Data Fusion:** The majority of systems are single-factor, reacting only to ambient light or a schedule. Even advanced platforms show data in silos, leaving it to a human operator to correlate that a traffic jam (from CCTV) and low visibility (from a weather feed) are happening in the same place. Our system automates this by implementing a Sensor Fusion module to create a single, holistic situational awareness score.
- **Static and Reactive Logic:** Existing systems use static, hard-coded IF-THEN rules. They are reactive and cannot adapt to novel situations or learn from past events. Our system is proactive, using an LLM-based Decision Engine that consults a RAG pipeline (Pinecone DB) to compare the current situation to historical incidents, allowing it to make more nuanced and adaptive decisions.
- **No Automated Safety Validation:** In current systems, if an operator or a faulty rule issues an unsafe command (e.g., dimming lights during a cyclone), the system will execute it. Our architecture includes a novel System Monitor (Agent 4), which acts as an independent LLM "Judge" to validate the safety and reasonableness of every decision before it is sent to the physical streetlights, providing a critical layer of automated trust.

- **Poor Handling of Anomalies:** Basic systems are easily fooled by anomalous data (e.g., a faulty sensor reading 0 lux at noon). Our pipeline includes a dedicated Anomaly & Behavior Detection module to identify and flag such data, preventing flawed information from corrupting the decision-making process.
- **"Black Box" Operations:** With most commercial platforms, the control logic is proprietary and hidden. Our system provides a transparent Admin Dashboard and a live Event Log that shows exactly why an AI decision was made (e.g., "Anomalies: High Congestion, Low Light") and whether the LLM Judge APPROVED it, building operator trust.

Chapter 3: Requirements for the Proposed System

Requirements Gathering

Requirements Gathering The requirements for the Smart Lighting Control System were gathered by analyzing core system architecture, including the multi-agent AI pipeline design, the structure of heterogeneous data inputs (CCTV, IoT, Weather), and the need for operator oversight and configuration control. The process involved defining input/output contracts between agents and establishing performance benchmarks critical for real-time urban management.

3.1 Functional Requirements

- **Weather Scenario Simulation:** Ability to trigger predefined weather conditions (e.g., `heavy_rainfall`, `cyclone_alert`) to test the AI agent's response.
- **Multi-sensor Data Collection:** Simultaneous fetching of real-time (or mock) weather, CCTV traffic, and IoT environmental data.
- **Automated Data Pipeline Execution:** End-to-end processing of collected data through Preprocessing, Sensor Fusion, and Anomaly Detection nodes.
- **AI-Driven Brightness Control:** Automatic calculation and application of optimal `brightness` levels to smart poles based on the Decision Engine's output.

- **Manual Pole Override:** Operator function to manually set `brightness` and lock control to bypass the AI agent's automated actions.
- **Configurable AI Thresholds:** Ability for operators to adjust key variables (e.g., `heat_threshold`, `congestion_threshold`) used by the Decision Engine logic.
- **Real-time State Distribution:** Push immediate updates of pole status and AI decisions to the frontend via WebSockets.
- **Agent Log Display:** Retrieval and display of a chronological log documenting all simulation triggers, agent decisions, and LLM Judge verdicts.

3.2 Non-Functional Requirements

- **Performance:** The end-to-end decision cycle (from trigger to control action) must be performed with low latency, generally targeting a few seconds to maintain real-time relevance.
- **Usability:** A clear, dark-mode, map-centric user interface (UI) allowing simple triggering of scenarios and instant visualization of changes.
- **Security:** Implementation of CORS middleware to restrict access to the FastAPI server only from trusted frontend origins.
- **Data Integrity:** Robust preprocessing and validation steps to handle and filter out noisy or anomalous sensor readings before fusion.
- **Scalability:** The backend architecture (FastAPI, LangGraph) must support the integration of more geographical zones and thousands of light poles.

3.3 Constraints

- **External API Dependency:** Reliance on the availability and data quality of external services (e.g., WeatherAPI, currently mocked) for environmental data input.
- **LLM Latency:** The inclusion of large language models (LLMs) in the Decision Engine and System Monitor introduces inherent latency, which must be managed to meet real-time performance targets.
- **RAG Infrastructure:** The functional requirement of RAG for decision support depends on a separate, available Pinecone VectorStore instance.

- Data Consistency: Ensuring logical consistency between disparate sensor data modalities (e.g., correlating CCTV congestion with IoT noise levels) is complex.

3.4 Hardware & Software Requirements

Hardware:

- Server/Host PC: For running the Python backend and Streamlit frontend.

Software:

- Backend Framework: Python (v3.10+), FastAPI.
- AI/LLM: LangGraph (for orchestration), Groq SDK (for Llama 3.1 model calls).
- Data/RAG: Pinecone Client, langchain-pinecone (Vector Store for historical incidents).
- Frontend: Next.js, React, TypeScript.
- Visualization/State: Leaflet/React-Leaflet (Map rendering), Redux/Redux Toolkit (State Management).
- Communication: Websockets, Uvicorn.

3.5 Techniques Utilized Till Date

- Multi-Agent LangGraph Pipeline: Orchestrating a complex workflow involving multiple specialized AI agents (Collection, Processing, Fusion, Decision, Monitor).
- Retrieval-Augmented Generation (RAG): Using vector similarity search (Pinecone) to retrieve historical incident data to ground the Decision Engine's recommendations.
- Heterogeneous Sensor Fusion: Combining processed weather data, CCTV traffic metrics, and IoT environmental metrics (AQI, noise, light) into a unified situational awareness score.
- LLM Validation / Monitoring: Using a second LLM (the "Judge") to critique and approve the core decision agent's suggested control action.
- Dynamic SVG Iconography: Utilizing custom SVG icons on the map that change appearance based on the current simulated weather state.

3.6 Tools Utilized Till Date

- Backend Framework: FastAPI, Uvicorn.
- AI Runtime: Groq, LangGraph, LangChain.
- Vector Database: Pinecone.
- Frontend: Next.js, React, Leaflet, React-Leaflet, Tailwind CSS.

- Development Tools: Visual Studio Code, TypeScript.

3.7 Project Proposal

An application should be developed to manage and optimize smart city lighting infrastructure by dynamically adjusting light pole brightness in response to real-time weather and traffic conditions. The system employs an AI-driven multi-agent pipeline built on LangGraph and Groq's Llama 3.1 to process heterogeneous sensor data (Weather, CCTV, IoT), calculate a comprehensive risk score, and generate autonomous control actions. The backend, built with FastAPI, manages the decision workflow, leverages RAG against a Pinecone vector store for historical context, and provides real-time state updates via WebSockets. The Next.js frontend features a central, dark-themed Leaflet map visualization with dynamic, weather-aware SVG icons and interactive controls, enabling operators to monitor system status, simulate scenarios, and configure AI parameters for enhanced energy efficiency and public safety.

Chapter 4: Proposed Design

4.1 System Design / Conceptual Architecture

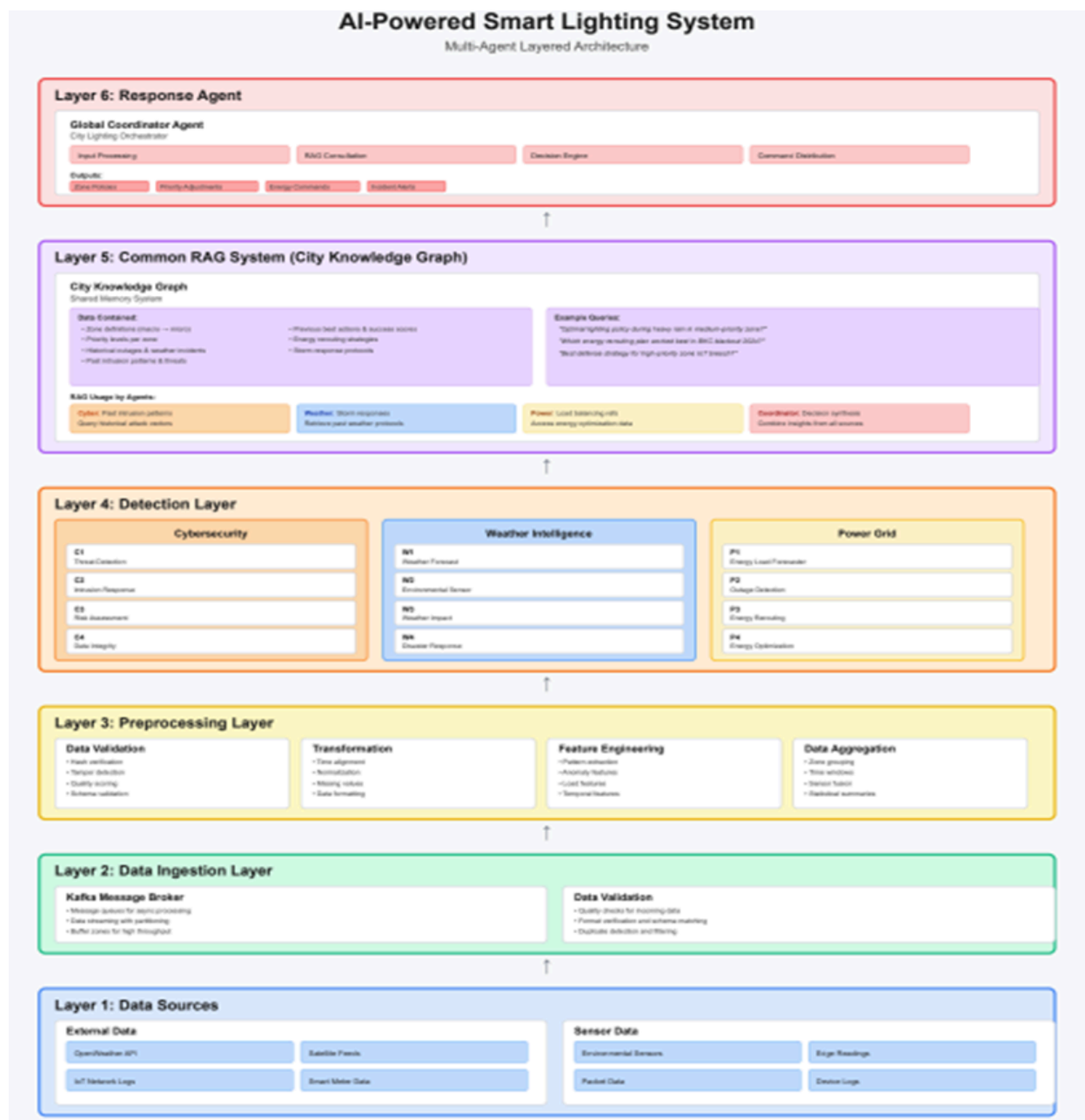


Fig. 4.1 Detailed System Architecture

The diagram depicts a sophisticated AI-Powered Smart Lighting System organized into six distinct layers, each providing essential functions to ensure intelligent and responsive street lighting management:

Layer 1: Data Sources gathers raw inputs including sensor data from streetlights (voltage, energy usage, environmental readings) and external data like weather and city events to provide a comprehensive data foundation.

Layer 2: Data Ingestion Layer uses a Kafka Message Broker for secure, high-throughput real-time ingestion of raw data streams, complemented by initial data validation to ensure input reliability.

Layer 3: Preprocessing Layer cleans, transforms, and enriches the raw data by engineering features that highlight critical patterns relevant for fault and outage detection.

Layer 4: Detection Layer applies AI algorithms to analyze the preprocessed data, specifically monitoring power grid health and weather conditions, and conducts cybersecurity checks to detect anomalies or outage events within milliseconds.

Layer 5: Common RAG System incorporates the city knowledge graph, enabling agents to contextualize faults using historical data and urban insights for informed decision-making.

Layer 6: Response Agent serves as the orchestrator that executes actions such as switching to backup power sources during outages and coordinates commands to maintain continuous lighting.

4.2 Block Diagram & Explanation

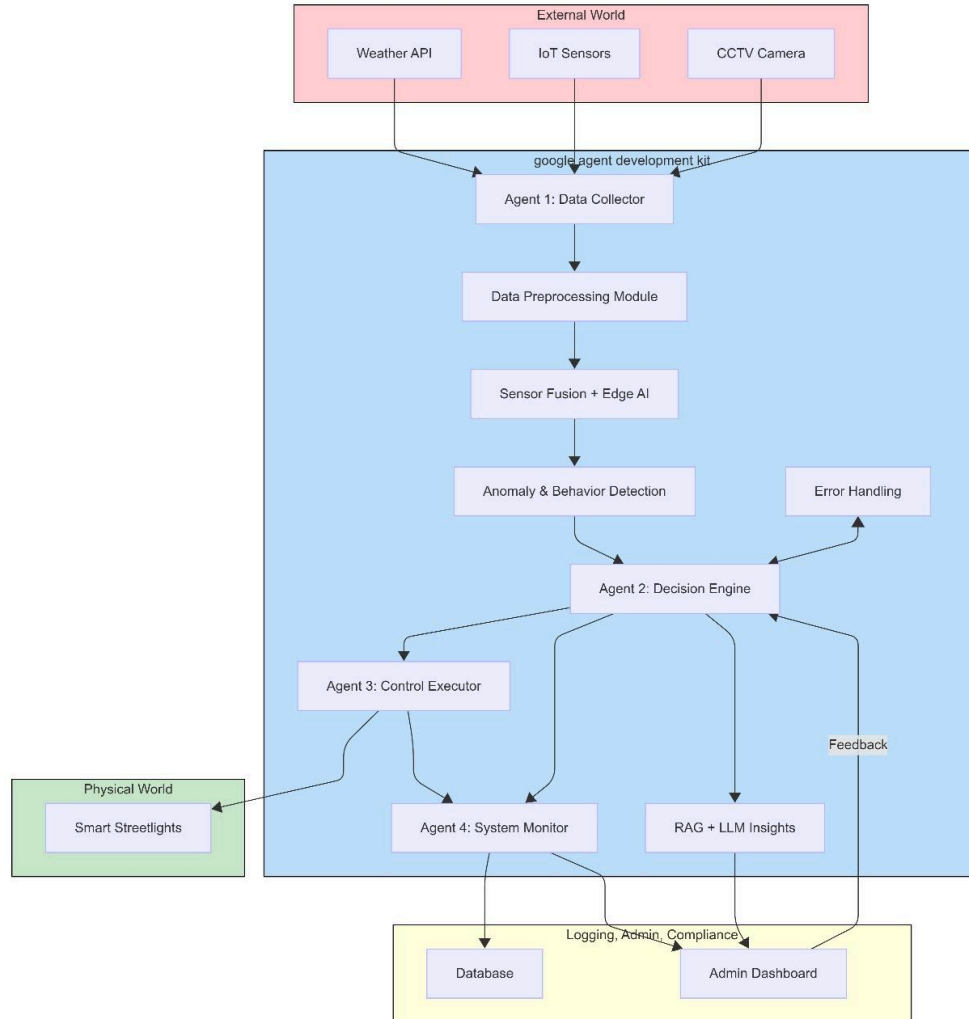


Fig. 4.2 Block Diagram

The system is structured into four main layers: External World, AI Core (Smart Agent Development Kit), Physical World, and Management/Database.

1. External World (Data Sources)

- This layer represents the raw inputs used to assess the city's current environment.
- Weather API: Provides meteorological data (e.g., temperature, condition).
- IoT Sensors: Provides environmental metrics (e.g., AQI, noise, ambient light).

- CCTV Camera: Provides traffic-related data (e.g., pedestrian/vehicle count, congestion level).

2. AI Core (Smart Agent Development Kit)

- This is the central nervous system, built as a multi-agent pipeline using frameworks like LangGraph and Groq.
- Agent 1: Data Collector: The entry point, responsible for fetching raw data from all external sources.
- Data Preprocessing Module: Cleans, validates, and standardizes the raw, heterogeneous data formats.
- Sensor Fusion + Edge AI: Merges data from different sensors into a unified, rich environmental state and calculates situational awareness scores.
- Anomaly & Behavior Detection: Applies rules and thresholds to identify unusual or hazardous conditions within the fused data (e.g., extreme heat, high congestion).
- RAG + LLM Insights: A retrieval-augmented generation component that consults historical incident data (e.g., Pinecone vector store) to provide contextual insights to the decision-making process.
- Agent 2: Decision Engine: The core logic node that assesses the anomalies and RAG insights to determine the optimal control action and operational recommendations.
- Agent 3: Control Executor: Translates the decision (e.g., "increase visibility lighting") into a specific, executable command (e.g., brightness: 90%) and applies it to the physical infrastructure.
- Agent 4: System Monitor: An independent "LLM Judge" that critiques the Decision Engine's output for safety, reasonableness, and effectiveness before the final action is executed.
- Error Handling: A mechanism to manage and log errors occurring throughout the pipeline, preventing system crashes.

3. Physical World (Output)

This represents the hardware being actively managed by the system.

- **Smart Streetlights:** The city's light poles, which receive control commands (e.g., brightness adjustments) from the Control Executor.

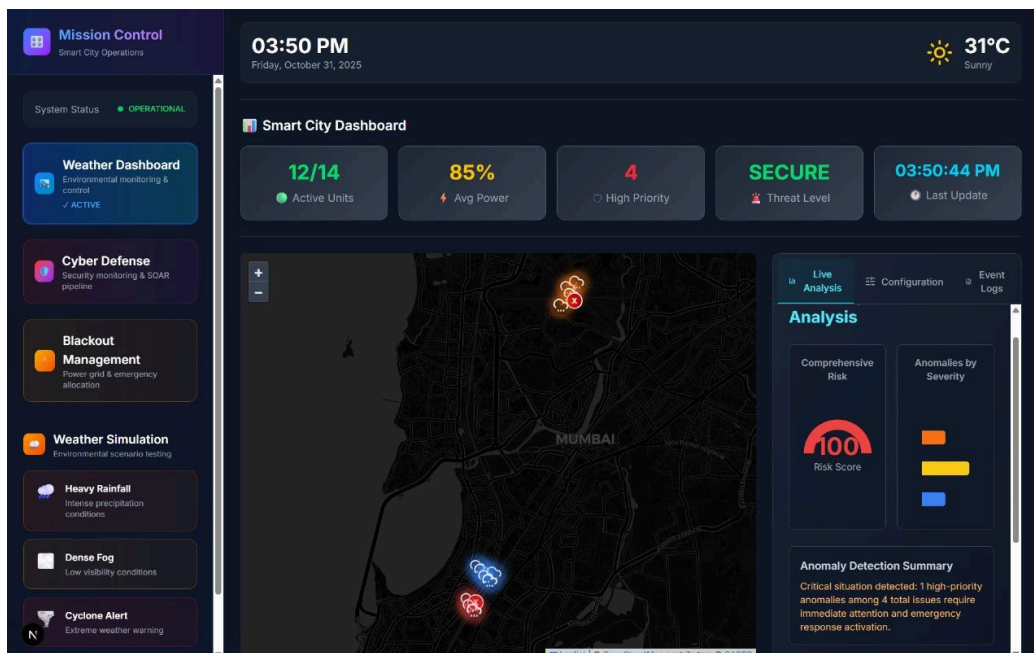
4. Management & Database Layer

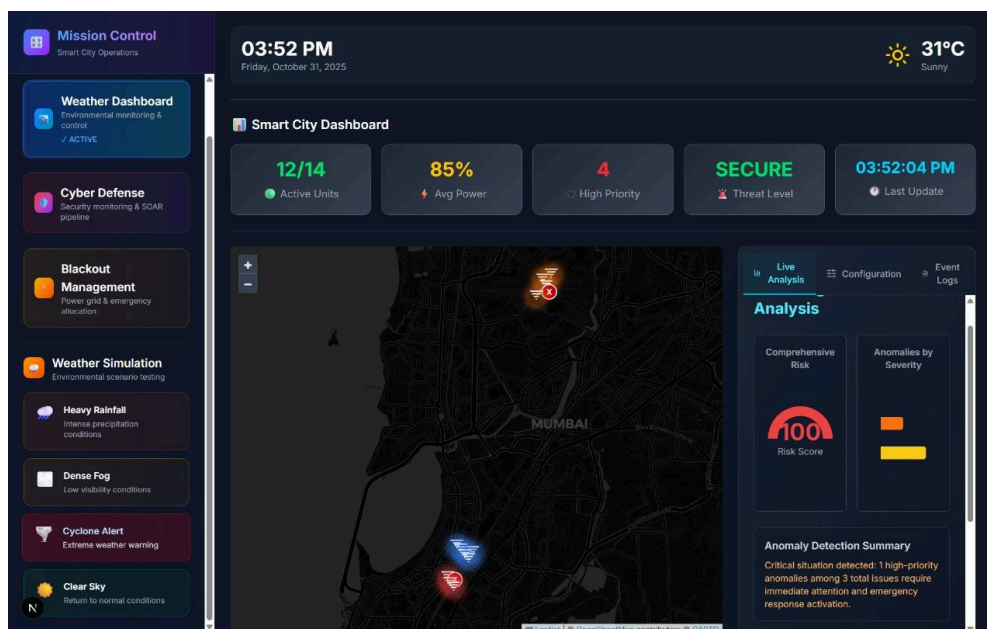
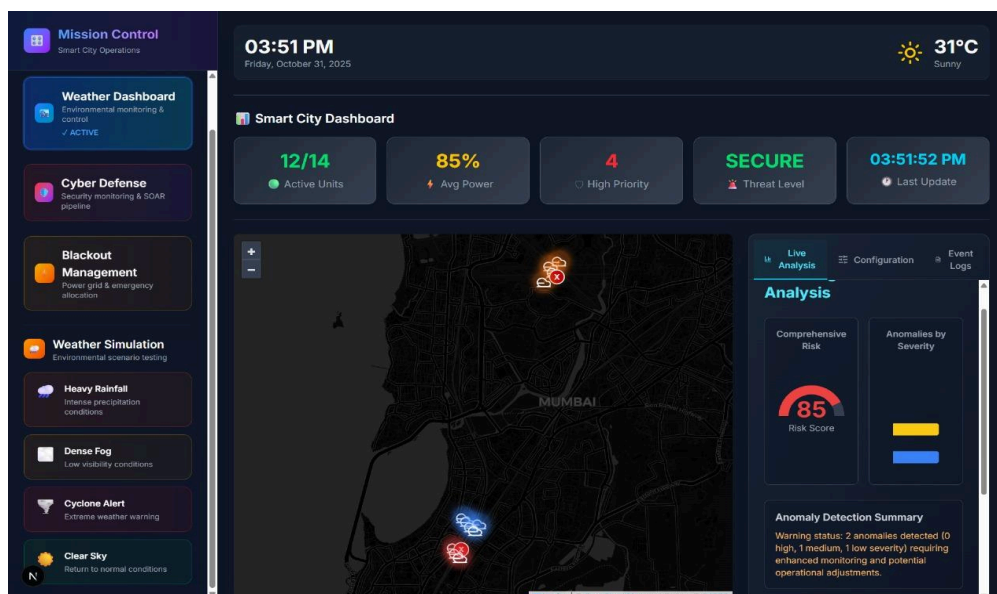
These components provide persistence, visibility, and management capabilities.

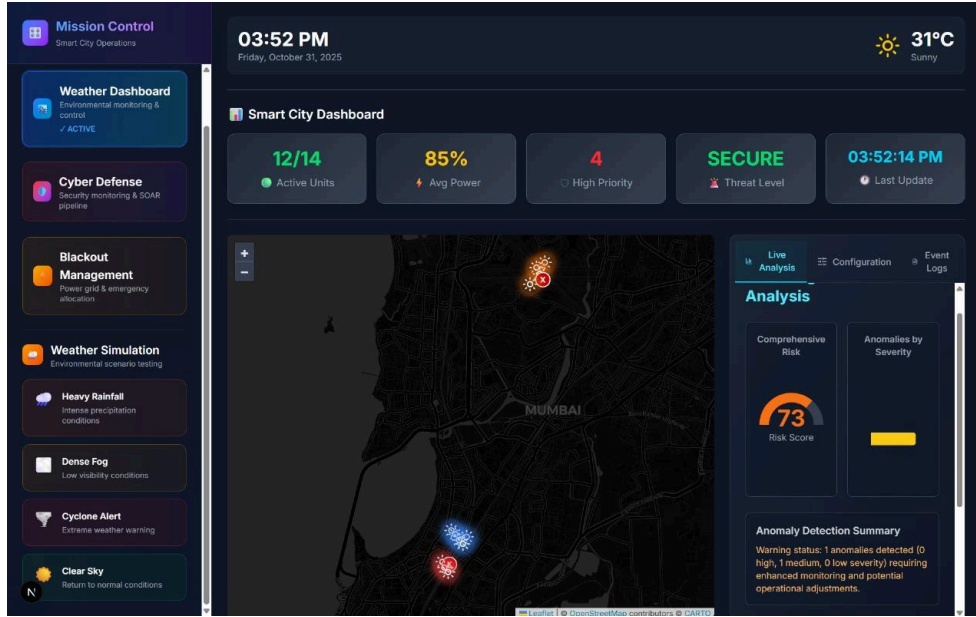
- **Database:** Stores logs, historical sensor readings, incident reports, and past control actions.
- **Admin Dashboard:** The primary user interface for monitoring system status, viewing logs, overriding controls, and configuring agent thresholds.
- **Logging, Admin, Compliance:** This represents the flow of audit trails, performance metrics, and compliance status updates to the dashboard for operator review.
- **Feedback:** Direct user feedback (e.g., manual overrides or configuration changes) can be fed back to the Decision Engine to improve future actions.

4.4 Detailed Design

Implementation Screenshots:







Chapter 5: Proposed Results & Discussions

5.1 Determination of Efficiency As the project is currently in the research and initial development stage, efficiency will be evaluated after full implementation. The primary metric will be the end-to-end pipeline latency, measured from the moment a scenario is triggered to the time the final control_action command is generated. The target is to achieve near-real-time response (under 3-5 seconds) to ensure the system can react swiftly to sudden environmental changes.

5.2 Determination of Accuracy The prototype's multi-agent pipeline correctly processes data through each node. Future testing will measure:

- **Decision Accuracy:** Comparing the Decision Engine's recommended brightness against a predefined matrix of "ideal" brightness levels for various anomaly combinations (e.g., heavy rain + high congestion should yield 90%+ brightness).
- **Validation Rate:** The percentage of decisions from Agent 2 (Decision Engine) that are APPROVED by Agent 4 (System Monitor), measuring the internal reliability of the AI logic.

5.3 Sensitivity Analysis Sensitivity analysis will be performed to observe the impact of data quality on the pipeline's final decision. This will involve injecting anomalous or missing data (e.g., "CCTV feed offline" or "IoT sensor data delayed") to test the robustness of the Sensor Fusion and Anomaly Detection modules in handling real-world data imperfections.

Chapter 6: Plan of Action for the Next Semester

6.1. Work Done Till Date

- 6.1. Work Done Till Date
 - Conducted detailed architectural design of the multi-agent system, defining the data flow from external sensors through the AI Core to the physical streetlights.
 - Gained critical insights into the operational challenges of urban infrastructure, particularly the need to correlate disparate data sources (weather, traffic, IoT) for holistic decisions.
 - Designed and developed the initial prototype, including:
 - Backend Server: A FastAPI application with WebSocket integration for real-time dashboard updates.
 - Frontend Dashboard: A Next.js application with Redux state management and an interactive Leaflet map displaying dynamic, weather-aware SVG icons.
 - Core AI Pipeline: A multi-node LangGraph agent (Collector, Preprocessing, Fusion, Decision, Monitor) using Groq Llama 3.1.
 - RAG Implementation: A functional Pinecone vector database seeded with historical incidents, ready for integration with the Decision Engine.
 - Simulation Controls: A UI sidebar for triggering predefined weather scenarios to test the full AI pipeline.

6.2. Plan for Project II

- Expand Prototype Functionality:
 - Implement the full, end-to-end RAG workflow, where the Decision Engine actively queries the Pinecone vector store to find similar historical incidents and uses that context to inform its final brightness recommendation.
 - Implement the operator-facing configuration panel, allowing AI decision thresholds (e.g., heat_threshold) to be adjusted from the UI and saved in the backend.
- Enhance Technical Capabilities:
 - Refine the Sensor Fusion algorithms to handle more complex correlations (e.g., wind speed vs. traffic density) and gracefully manage missing data from one or more sources.
 - Optimize the Groq/LangGraph pipeline to minimize LLM-related latency and ensure decisions are made within the target real-time window.
- Field Expansion (Simulation):
 - Conduct a large-scale simulation modeling all three city zones with their full complement of poles.
 - Run a battery of tests using all weather scenarios against different configurations to gather quantitative data on projected energy savings vs. safety/visibility metrics.
- System Refinement and Testing:
 - Enhance the UI/UX by implementing the Live Clock & Weather widget and the Zone Details Panel (activated on map click).

- Perform comprehensive integration testing, analyze operator feedback from the simulation panel, and debug the end-to-end system before a hypothetical deployment.

Chapter 7: Conclusions

The project, AI - Powered Weather Intelligence for Smart Lighting, aimed to address the critical issues of urban energy inefficiency and public safety risks by developing an autonomous, AI-driven control system for smart city infrastructure.

Through the architectural design process, it was found that traditional, single-sensor, or schedule-based control logic is insufficient for managing the complex and dynamic environment of a modern city. The developed prototype successfully demonstrates how a multi-agent AI pipeline can solve this by integrating:

- Heterogeneous Sensor Fusion to combine data from weather, CCTV, and IoT sources into a single, actionable insight.
- RAG-based Decision Making to leverage historical incident data from a vector store, ensuring decisions are context-aware and not just rule-based.
- An LLM-based Validation Layer (System Monitor) to act as a crucial safety check on all automated decisions, enhancing system trustworthiness.

This approach not only promotes significant energy savings but also enhances urban resilience and public safety by adapting proactively to hazardous conditions. It encourages responsible infrastructure management and aligns with national Smart City and sustainability goals.

While challenges remain in minimizing LLM latency and ensuring the reliability of real-world sensor data, the prototype outcomes confirm the potential of this agentic system to serve as a robust foundation for next-generation urban infrastructure management.

The next phase will focus on optimizing the pipeline's performance, fully implementing the RAG-based decision-making loop, and conducting larger-scale simulations to validate the system's real-world impact on energy consumption and safety.

Chapter 8: References

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